General Bite #8: Test Equipment for the Ham Station

Oscilloscope, multimeter (digital vs analog), two-tone test, directional wattmeter, antenna analyzer.

Group G4B · 13 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

The Oscilloscope

Question	Answer	Q
Test instrument with horizontal and vertical channel amplifiers	**An oscilloscope** (G4B01).	G4B01
Oscilloscope advantage vs DVM	**Complex waveforms can be measured** (G4B02).	G4B02
Best instrument for checking a CW keying waveform	**An oscilloscope** (G4B03). Shows the rise/fall/key clicks.	G4B03
Signal source connected to vertical input for RF envelope	**The attenuated RF output of the transmitter** (G4B04). Direct RF is way too hot for the scope input.	G4B04

Multimeters

Question	Answer	Q
Why do voltmeters have high input impedance?	**It decreases the loading on circuits being measured** (G4B05). Low-Z meter would pull current from the circuit and falsify the reading.	G4B05
Digital multimeter advantage vs analog	**Higher precision** (G4B06). Digit count beats needle-on-scale resolution.	G4B06
When is an analog multimeter preferred?	**When adjusting circuits for maximum or minimum values** (G4B09). The swinging needle peaks/dips are easier to follow than digit flicker.	G4B09

The Two-Tone Test

What signals are used for a two-tone test? Two non-harmonically related audio signals (G4B07).
What does a two-tone test analyze? Linearity of a transmitter (G4B08). Two clean tones in; if the transmitter is linear, only those two tones come out. Any additional products on a spectrum analyzer – particularly third-order products like 2F1-F2 near the originals – indicate intermodulation distortion (IMD). Any extra products visible on a spectrum analyzer indicate intermodulation distortion (IMD) – the transmitter is being driven non-linearly.

Directional Wattmeter

What can be determined with a directional wattmeter? Standing wave ratio (G4B10). Forward and reflected power readings let you compute SWR.

Antenna Analyzer

Question	Answer	Q
What must be connected to an antenna analyzer for SWR measurements?	**Antenna and feed line** (G4B11).	G4B11
Effect of strong nearby transmitter signals on an antenna analyzer	**Received power that interferes with SWR readings** (G4B12). The analyzer's tiny test signal gets swamped by big nearby RF.	G4B12
What can an antenna analyzer measure?	**Impedance of coaxial cable** (G4B13). Also resonant frequency, SWR vs frequency, complex impedance.	G4B13

Mental Cheat Sheet

- Oscilloscope = vertical and horizontal channel amps.
- Scope advantage = sees complex waveforms.
- Best for CW keying waveform: **scope**.
- RF envelope check: feed **attenuated TX output** to vertical input.
- Voltmeter high Z = low loading on circuit.
- DMM advantage = higher precision.
- Analog meter wins for peak/null tuning.
- Two-tone test: **two non-harmonic AF tones**; analyzes **linearity**.
- Directional wattmeter → SWR.
- Antenna analyzer needs **antenna + feed line** connected.
- Strong nearby TX **interferes with** SWR readings.
- Antenna analyzer can measure coax impedance.

Common Traps

- Scope advantage is **complex waveforms**, not "less power" or "greater precision."
- For CW keying you want a **scope**, not a sidetone monitor or wavemeter.
- Voltmeter Z is high to **not load** the circuit, not to "measure higher voltages."
- Two-tone test uses **AF tones, non-harmonic**; tests **linearity** (IMD), not carrier suppression.
- Directional wattmeter measures **SWR** via fwd/refl readings, not "front-to-back."
- Antenna analyzer interference from nearby TX shows up as SWR-reading instability.

- Don’t connect just a transmitter or receiver to an antenna analyzer for SWR – need the actual antenna and feed line.

Takeaway: Scope sees waveforms (use it for CW keying, RF envelope). DMM has precision; analog meter has peak-and-null feel. Two-tone test = linearity test (two non-harmonic AF tones into the transmitter, watch the output spectrum). Directional wattmeter gives SWR via fwd/refl readings. Antenna analyzer measures impedance/SWR – connect antenna and feed line; watch for nearby-transmitter interference.

Bite 8 Question Bank (G4B)

G4B01. What item of test equipment contains horizontal and vertical channel amplifiers?
A. An ohmmeter.
B. A signal generator.
C. An ammeter.
D. An oscilloscope.

G4B02. Which of the following is an advantage of an oscilloscope versus a digital voltmeter?
A. An oscilloscope uses less power.
B. Complex impedances can be easily measured.
C. Greater precision.
D. Complex waveforms can be measured.

G4B03. Which of the following is the best instrument to use for checking the keying waveform of a CW transmitter?
A. An oscilloscope.
B. A field strength meter.
C. A sidetone monitor.
D. A wavemeter.

G4B04. What signal source is connected to the vertical input of an oscilloscope when checking the RF envelope pattern of a transmitted signal?
A. The local oscillator of the transmitter.
B. An external RF oscillator.
C. The transmitter balanced mixer output.
D. The attenuated RF output of the transmitter.

G4B05. Why do voltmeters have high input impedance?
A. It improves the frequency response.
B. It allows for higher voltages to be safely measured.
C. It improves the resolution of the readings.
D. It decreases the loading on circuits being measured.

G4B06. What is an advantage of a digital multimeter as compared to an analog multimeter?
A. Better for measuring computer circuits.
B. Less prone to overload.
C. Higher precision.
D. Faster response.

G4B07. What signals are used to conduct a two-tone test?
A. Two audio signals of the same frequency shifted 90 degrees.
B. Two non-harmonically related audio signals.
C. Two swept frequency tones.
D. Two audio frequency range square wave signals of equal amplitude.

G4B08. What transmitter performance parameter does a two-tone test analyze?
A. Linearity.
B. Percentage of suppression of the carrier and undesired sideband for SSB.
C. Percentage of frequency modulation.
D. Percentage of carrier phase shift.

G4B09. When is an analog multimeter preferred to a digital multimeter?
A. When testing logic circuits.
B. When high precision is desired.
C. When measuring the frequency of an oscillator.
D. When adjusting circuits for maximum or minimum values.

G4B10. Which of the following can be determined with a directional wattmeter?
A. Standing wave ratio.
B. Antenna front-to-back ratio.
C. RF interference.
D. Radio wave propagation.

G4B11. Which of the following must be connected to an antenna analyzer when it is being used for SWR measurements?
A. Receiver.
B. Transmitter.
C. Antenna and feed line.
D. All these choices are correct.

G4B12. What effect can strong signals from nearby transmitters have on an antenna analyzer?
A. Desensitization which can cause intermodulation products which interfere with impedance readings.
B. Received power that interferes with SWR readings.
C. Generation of harmonics which interfere with frequency readings.
D. All these choices are correct.

G4B13. Which of the following can be measured with an antenna analyzer?
A. Front-to-back ratio of an antenna.
B. Power output from a transmitter.
C. Impedance of coaxial cable.
D. Gain of a directional antenna.

Answer Key

G4B01: D	G4B02: D	G4B03: A	G4B04: D
G4B05: D	G4B06: C	G4B07: B	G4B08: A
G4B09: D	G4B10: A	G4B11: C	G4B12: B
G4B13: C			